

120 Days of Quantitative Finance

Month 4: Stochastic Calculus

Week 13: Brownian Motion

Day 90: Simulating Brownian Motion

1 Introduction

Brownian Motion is a continuous-time stochastic process widely used in financial modeling.

Since continuous processes cannot be simulated exactly, we approximate them using discrete time steps.

2 Discrete Approximation

Brownian Motion is approximated as:

$$W_{t+\Delta t} = W_t + \sqrt{\Delta t} \cdot \epsilon_t$$

where:

$$\epsilon_t \sim \mathcal{N}(0, 1)$$

3 Algorithm

1. Choose time horizon T and number of steps N
2. Compute $\Delta t = \frac{T}{N}$

3. Generate random increments:

$$dW_t = \sqrt{\Delta t} \cdot \epsilon_t$$

4. Construct path:

$$W_t = \sum dW_t$$

4 Statistical Properties

Simulated Brownian Motion satisfies:

$$E[W_t] = 0$$

$$Var(W_t) = t$$

5 Financial Interpretation

Brownian Motion models:

- unpredictable price movements
- accumulation of random shocks
- increasing uncertainty over time

6 Multiple Paths

To understand randomness, we simulate multiple paths:

- each path represents a possible market trajectory
- divergence increases over time due to variance growth

7 Connection to Finance

Brownian Motion is used in:

- Geometric Brownian Motion (GBM)

- Black-Scholes model
- Stochastic Differential Equations

8 Conclusion

Simulating Brownian Motion provides intuition for how randomness evolves over time.

It forms the foundation for modeling asset prices in continuous time.